

Hemanth Kumar Galam

Robotics Software Engineer | AI & Machine Learning

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PROFESSIONAL SUMMARY

Robotics Software Engineer combining 5+ years of robust backend distributed systems experience with 2+ years of applied robotics research. Specializes in ROS2, Sim2Real workflows (NVIDIA Isaac Sim), and optimizing edge AI inference (TensorRT) to architect low-latency perception systems for Human-Robot Collaboration.

TECHNICAL SKILLS

Robotics & ROS: ROS2, C++ ROS API, NVIDIA Isaac Sim, NVIDIA Omniverse, RViz, Sim2Real

AI/ML & Vision: PyTorch, TensorFlow, TensorRT, GCNs, Transformers, Edge AI Inference

Languages: C++, Python, JavaScript, TypeScript

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, WebSockets, RabbitMQ

Databases & Caching: PostgreSQL, MongoDB, Redis

Cloud & DevOps: Docker, Kubernetes, AWS (Lambda, S3, RDS), GitHub Actions, CI/CD, Linux, Git

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant -- ML & Robotics Engineer

Jan 2024 -- Mar 2026

Fraunhofer IPA

Stuttgart, Germany

- Generated synthetic datasets in **NVIDIA Isaac Sim** and trained GCN/Transformer models, achieving **87% accuracy** on human action recognition.
- Boosted Sim2Real transfer accuracy from **68% to 84%** through domain randomization and targeted synthetic-to-real data augmentation.
- Reduced inference latency by **88% (125 ms to 15 ms)** using **TensorRT** optimization on Jetson Orin Nano, integrated directly into a ROS2 package.
- Architected and Dockerized a real-time ROS2 perception pipeline streaming skeleton telemetry to an Angular UI via WebSockets with **<100 ms latency**.
- Evaluated OpenVLA foundation models on edge compute nodes to benchmark real-time inference feasibility for industrial Human-Robot Collaboration.

Software Engineer Intern

Apr 2023 -- Sep 2023

Rohde & Schwarz GmbH

Munich, Germany

- Developed a custom RL-encoding compression package in JavaScript, reducing map-loading latency by **40%**.
- Evaluated Autoencoders for map tile compression by preparing custom spatial datasets, achieving a **60% reduction** in tile size.
- Maintained **95% test coverage** using Jest within an Agile/Scrum team structure.

Senior Software Engineer

Dec 2020 -- Apr 2023

Ensar Solutions Pvt. Ltd.

Hyderabad, India

- Designed high-concurrency access management architecture for **200,000+ daily active users** across 40+ REST APIs.
- Optimized PostgreSQL queries and indexing strategies, reducing API response times by **40%**.

PROJECTS

Sim2Real HRC Perception Pipeline (ROS2 & Isaac Sim)

- Built an end-to-end synthetic data generation pipeline in NVIDIA Isaac Sim, training GCNs & Transformers for real-time edge action recognition on Jetson Orin hardware.

Migrator ETL Tool

- Architected a scalable ETL tool for high-integrity, zero-loss data migration between SQL (PostgreSQL) and NoSQL (MongoDB) databases.

Multi-LLM FastAPI & Model Router

- Built a high-performance FastAPI backend integrating multiple open-source LLMs via a unified interface with optimized prompt routing.

EDUCATION

Master of Science in Electrical and Computer Engineering

Oct 2022 -- Aug 2026

University of Stuttgart, Germany

- **Master Thesis (Grade: 1.3):** "Domain-Specific Action Recognition for HRC using Synthetic Data."
- Developed a real-time edge perception pipeline using GCN and Transformers for industrial collaboration.
- **Research Thesis (Grade: 1.0):** "ML-based Predictive Modeling for Thermal Runaway."
- Integrated Regression, Classification, and YOLO-based detection for battery safety monitoring.
- **Focus:** Machine Learning and Deep Learning (GPA: 2.1 / Good).

Bachelor of Technology in Electrical and Electronics Engineering

Jul 2017 -- Mar 2021

Jawaharlal Nehru Technological University, India

ACHIEVEMENTS

Winner -- sustainATHON 2023 (Stuttgart, Germany) •

3rd Place -- Bayerwald Hackathon (Deggendorf, Germany)

LANGUAGES

German (A2), English (C1 - Fluent)